SAWL-Net: Statistical Attentions & Wavelet-Aided Lightweight Network

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Motivation

Problem Context:

- Histopathological images contain complex textural and structural information. They exhibit *high inter-class similarity* and *large intra-class variance*, leading to failure of conventional CNNs to capture **statistical dependencies** among feature channels.

$$\textbf{Objective: } f_\theta(X) \rightarrow y, \quad \text{minimize } \mathcal{L}(f_\theta(X), y) \text{ while enforcing}$$
$$\underbrace{\text{Statistical Regularity}}_{\text{Channel-wise correlation}} + \underbrace{\text{Spectral Locality}}_{\text{Wavelet domain refinement}} .$$

Framework Overview:

- **Backbone:** MobileNetV2.
- **Statistical Attention:** leverages Pearson and Spearman correlations.
- **Wavelet Conv1D:** extracts localized frequency-domain patterns.

Goal: A lightweight yet statistically consistent network that integrates feature correlation and spectral awareness for histopathological classification.

Mathematical Formulation

Feature Representation:

$$FM = \{FM_1, \ldots, FM_C\}, \qquad \mu = \frac{1}{C}\sum_{i=1}^{C} FM_i$$

Statistical Correlations:

$$\mathrm{PCC}(\mu, FM_i) = \frac{\sum_j (\mu_j - \bar{\mu})(FM_{i,j} - \overline{FM_i})}{\sqrt{\sum_j (\mu_j - \bar{\mu})^2}\sqrt{\sum_j (FM_{i,j} - \overline{FM_i})^2}}$$

$$\rho_{SRC} = 1 - \frac{6\sum_j (r_{\mu_j} - r_{FM_{i,j}})^2}{n(n^2-1)}$$

Fusion & Mapping:

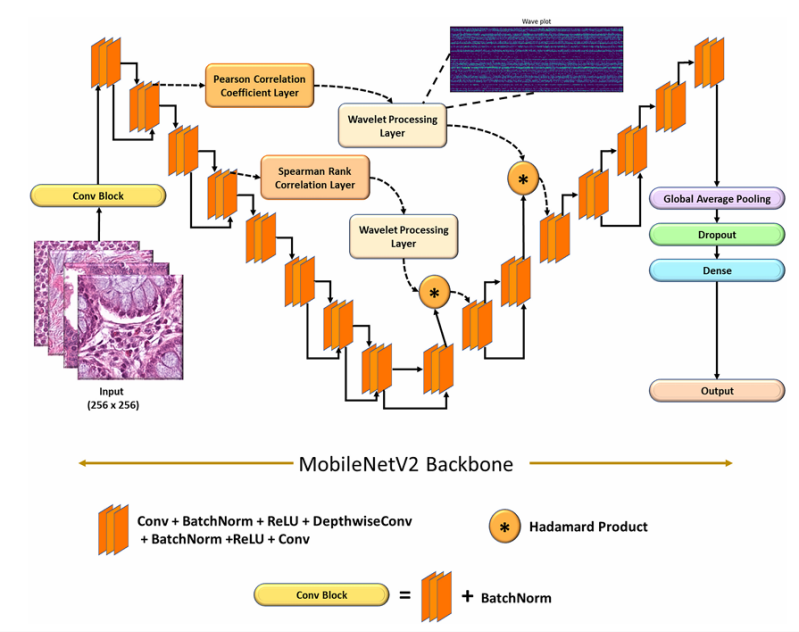
$$S_i = \alpha\,\mathrm{Corr} + (1-\alpha)\,\mathrm{CS}, \quad w_i = e^{a(S_i - h)}c + k, \quad FM_i' = w_i \cdot FM_i$$

Wavelet Processing: Flatten $FM_i' \Rightarrow$ Conv1D $\Rightarrow$ Local Patterns $\Rightarrow$ Fused via Hadamard Product

$$F_{out} = F_{att} \odot F_{wave}$$

SAWL-Net: High-Level Architecture

- **Backbone:** MobileNetV2
- **Two Statistical Attention:**
  - **Outer Path:** Pearson Correlation + Cosine Similarity: emphasizes *low-level texture features*.
  - **Inner Path:** Spearman Rank + Cosine Similarity: refines *high-level semantic representations*.
- **Spectral Refinement:** Wavelet-inspired Conv1D: enforces local smoothness across channels.

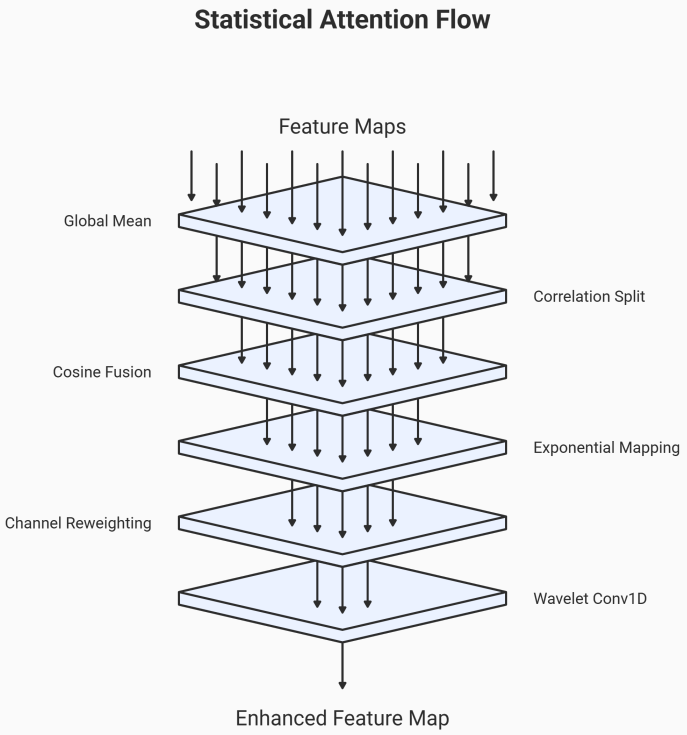


(SAWL-Net schematic)

Statistical Attentions Mechanism

Working Principle:

- Identifies channels statistically correlated with the global feature map μ.
- Computes dependencies via Pearson, Spearman, and Cosine similarity.
- High-correlation channels receive amplified weights w_i; weak or noisy ones are dropped.

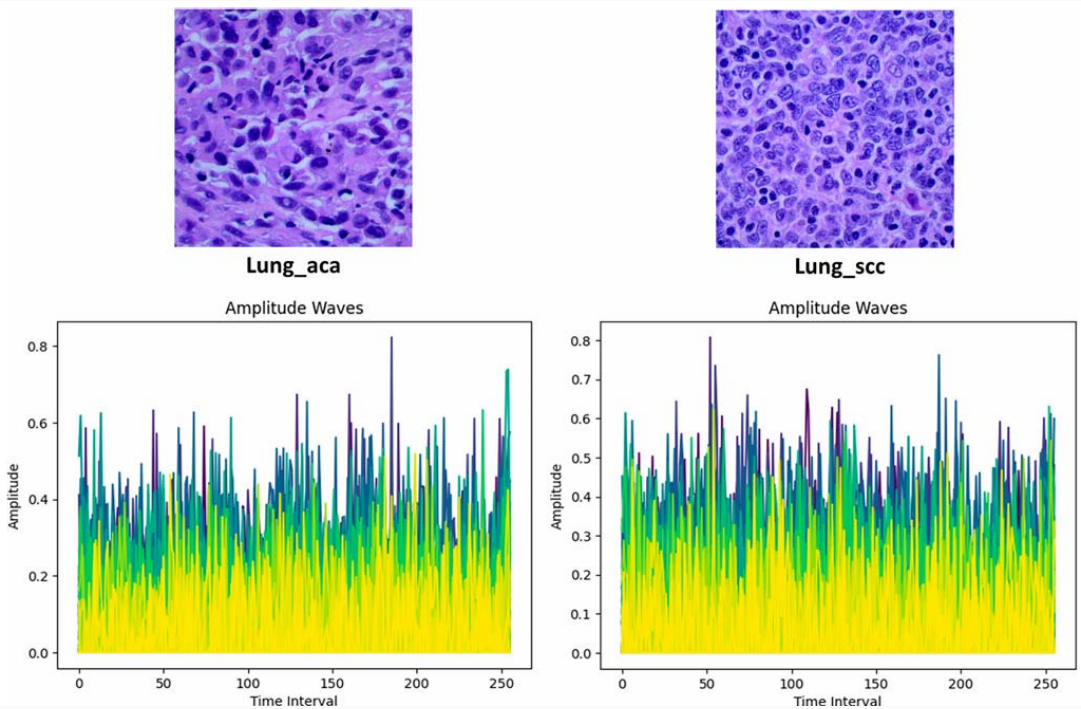


Flowchart showing the operations on feature maps.

Wavelet-Inspired Cross-Channel Processing

Concept:

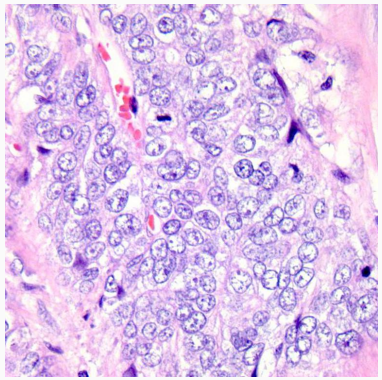
- Converts channel features into 1D sequences, for frequency-based filtering.
- Conv1D kernel captures inter-channel dependencies.
- Combines with attention output via Hadamard fusion.



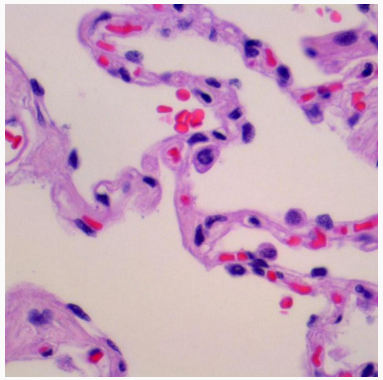
Sample images of two lung classes and their corresponding amplitude wave maps; the data of each channel is treated as amplitude and plotted.

Dataset & Training Protocol

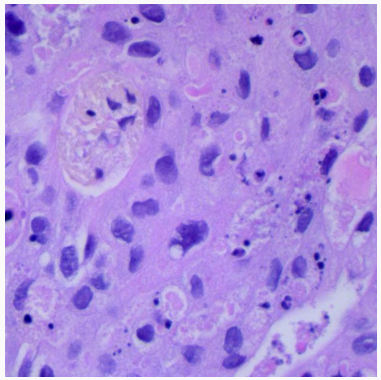
Datasets: LC25000 (Lung & Colon) and BreakHis (Breast Histopathology)



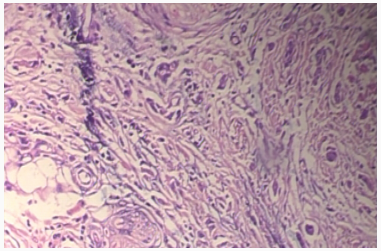
Colon Adenocarcinoma



Lung Normal



Lung SCC



Breast Malignant

Training Configuration:

- Optimizer: Adam ($\beta_1 = 0.9, \beta_2 = 0.999$), LR $= 1 \times 10^{-5}$.
- Batch size $= 16$, Epochs $= 50$.
- Loss: Sparse Categorical Cross-Entropy.
- Metrics monitored: Accuracy, Precision, Recall, and F1-score.

Both datasets used to evaluate SAWL-Net's generalization across organ-specific histopathological domains.

Input, Output & Trainable Parameters

Input:

- RGB histopathology patches of size 256×256.
- Images resized to 256×256, normalized.
- Stratified 70/10/20 split for train/validation/test.

Output:

- Softmax vector of class probabilities:

$$y = [P(\text{colon_aca}), P(\text{colon_n}), P(\text{lung_aca}), P(\text{lung_n}), P(\text{lung_scc})]$$

- Highest probability corresponds to predicted class.

Trainable Parameters:

- MobileNetV2 backbone weights (depthwise + pointwise convolution).
- 1D convolution filters in wavelet layer.

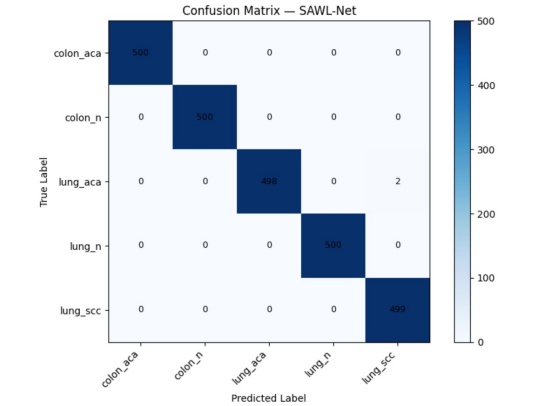
Statistical Attention block is non-trainable. It computes correlations analytically thus improving interpretability and reducing parameters.

Experimental Results and Analysis: LC25000

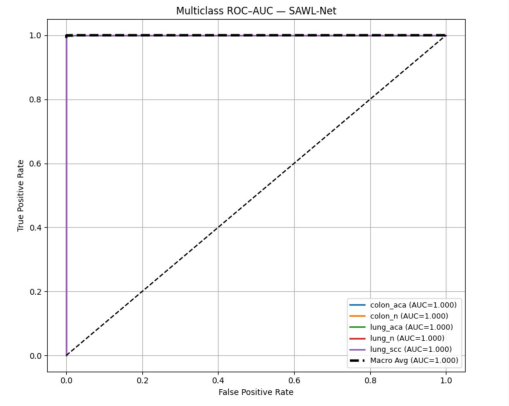
LC25000 Dataset: Sample patches and evaluation visuals

```
Classification report:
              precision    recall  f1-score   support

   colon_aca       1.00      1.00      1.00       500
     colon_n       1.00      1.00      1.00       500
    lung_aca       1.00      1.00      1.00       500
      lung_n       1.00      1.00      1.00       500
    lung_scc       1.00      1.00      1.00       499

    accuracy                           1.00      2499
   macro avg       1.00      1.00      1.00      2499
weighted avg       1.00      1.00      1.00      2499

Confusion matrix:
 [[500   0   0   0   0]
 [  0 500   0   0   0]
 [  0   0 498   0   2]
 [  0   0   0 500   0]
 [  0   0   0   0 499]]
ROC-AUC (macro OVR): 0.9999893886823171
```

Classification report

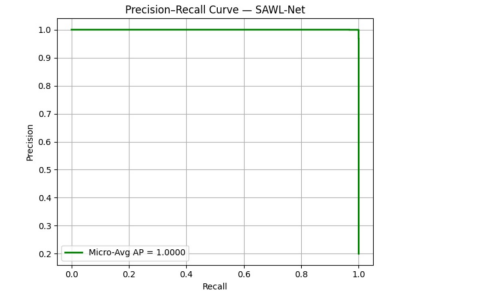


Confusion Matrix



ROC curve

```
Per-Class AUC Values:
  colon_aca              : 1.0000
  colon_n                : 1.0000
  lung_aca               : 1.0000
  lung_n                 : 1.0000
  lung_scc               : 1.0000

Macro-average ROC-AUC: 1.0000
```



Precision Recall Curve

ROC–AUC (Macro OVR):

0.999996

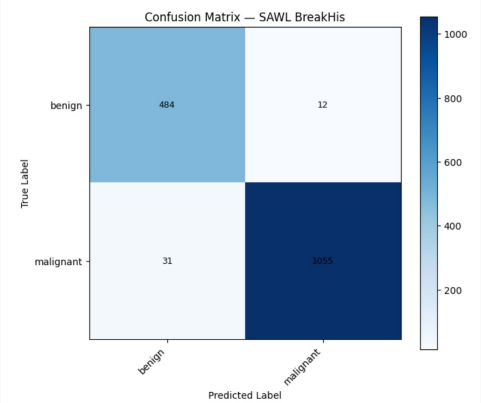
Observation: Near-perfect performance across all classes with minimal confusion between `lung_scc` and `lung_aca`.

Experimental Results and Analysis: BreaKHis

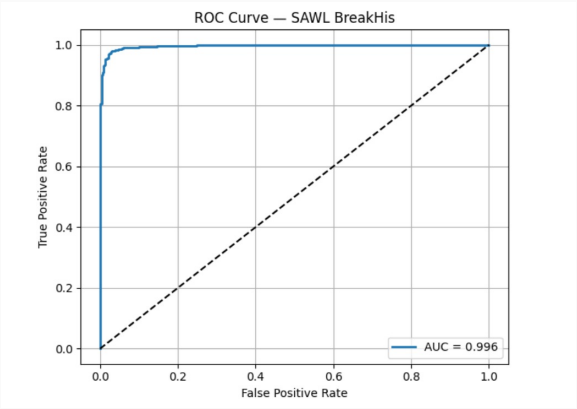
BreaKHis Dataset: Sample patches and evaluation visuals

```
For 50 epochs
Classification report:
              precision    recall  f1-score   support

      benign       0.92      0.98      0.95       496
   malignant       0.99      0.96      0.98      1086

    accuracy                           0.97      1582
   macro avg       0.95      0.97      0.96      1582
weighted avg       0.97      0.97      0.97      1582

Confusion matrix:
 [[ 486   10]
 [  43 1043]]
ROC-AUC (macro OVR): 0.9957152616883502
```

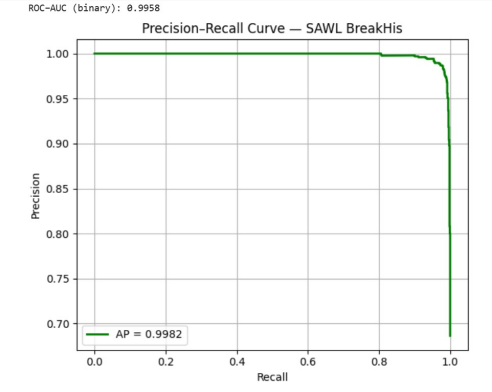
Classification report (summary)



Confusion Matrix (visual)



ROC curve (macro)



Precision Recall Curve

ROC–AUC (Macro OVR):

0.9957

Observation: High recall on malignant class; false-negative cases (43) indicate scope for improved stain-robust augmentations.

Selective Monte–Carlo Δ-Addition (Selective MC)

Goal: uncertainty-aware refinement, perform Monte-Carlo Dropout only on low-confidence samples.

Formulation:

$$\bar{p} = \frac{1}{T}\sum_{t=1}^{T} p^{(t)}, \qquad \mathrm{MI} = H(\bar{p}) - \frac{1}{T}\sum_{t} H(p^{(t)}), \quad p_{\text{corr}} = \text{sharpen}(\bar{p};\alpha)$$

```
Improved MC Δ-Addition: 100%|▰▰▰
```

Evidential Head — Dirichlet-Based Uncertainty Modeling

Goal: Model both prediction confidence and epistemic uncertainty within a single forward pass using a Dirichlet evidential head.

Concept:

- Replaces the final softmax layer with a lightweight head predicting Dirichlet parameters $\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \ldots, \alpha_K]$.
- Each $\alpha_k > 0$ represents the *evidence* supporting class k.
- The Dirichlet distribution encodes a distribution *over softmax vectors*, combining belief and uncertainty.

Mathematical Formulation:

$$p(\mathbf{p} \mid \boldsymbol{\alpha}) = \frac{1}{B(\boldsymbol{\alpha})} \prod_{k=1}^{K} p_k^{\alpha_k - 1}, \quad \hat{p}_k = \frac{\alpha_k}{S}, \quad S = \sum_{k=1}^{K} \alpha_k$$

$$\text{Vacuity (ignorance)}: \ v = \frac{K}{S}, \quad \text{Belief mass}: \ b_k = \frac{\alpha_k - 1}{S}$$

Frame Title

Interpretation:

- Large $\alpha_k \Rightarrow$ strong evidence, confident belief.
- Small $\alpha_k \Rightarrow$ weak evidence, high vacuity (uncertainty).
- Enables single-pass uncertainty estimation without Monte Carlo sampling.

```
Evidential Δ-Addition (selective): checked 131 accepted/refined 5
Accuracy before: 0.972819216182048 after: 0.9721871049304678 delta: -0.0006321112515802474
Saved changed samples list to results/figs_breakhis/evidential_delta_changed.csv
Classification report (after evidential Δ):
              precision    recall  f1-score   support

      benign     0.9431    0.9698    0.9563       496
   malignant     0.9860    0.9733    0.9796      1086

    accuracy                         0.9722      1582
   macro avg     0.9646    0.9715    0.9679      1582
weighted avg     0.9726    0.9722    0.9723      1582
```

(a)Evidential –Addition: Post-Uncertainty Refinement Performance.

Evidential Head transforms deterministic logits into a Dirichlet belief, providing calibrated confidence and uncertainty directly from model evidence.

Conclusion

Conclusion:

- The proposed SAWL-Net effectively integrates **statistical attention** and **wavelet-driven channel refinement** for accurate and interpretable histopathological classification.
- Incorporating **Selective Monte–Carlo delta Addition** and **Evidential–Conformal routing** provides a robust uncertainty-aware framework, maintaining accuracy while improving calibration and efficiency.
- The entire pipeline remains lightweight, supporting real-time inference on resource-limited devices.

Thank You!